Research Method of Data Science

Week 11: Bivariate Analysis and Inference

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Tuesday agenda

Focus: choose evidence that fits the question and explain what it supports. Companion notebook: **W11_RMDS_Inference_Lab.ipynb**

1. Comparing two variables with real evidence
2. Sampling variation and confidence intervals
3. Tests, null models, and p-values
4. Effect size and substantive importance
5. Evidence choices for research questions
6. Preparation for Thursday

Description and inference

Description

A graph or table summarises the records you observed.

Inference

The analysis asks what those records support about a defined target under explicit assumptions.

PAIR DISCUSSION · 3 MIN

If two groups have different average earnings, what else must you know before claiming that a programme worked?

1. Bivariate questions have different forms

Group + number	Training group and earnings	Group summaries and distributions
Number + number	Earnings in two baseline years	Scatterplot and association measure
Group + binary	Training group and positive earnings	Counts, proportions, and percentage-point gap

> Variable types help choose the comparison. Study design determines the claim.

Study question and observed earnings

National Supported Work study

Primary question: How do mean 1978 real earnings differ between the training and control groups?

The analysis uses the source-dollar scale. The sample does not represent all workers or current monetary values.

Observed participants

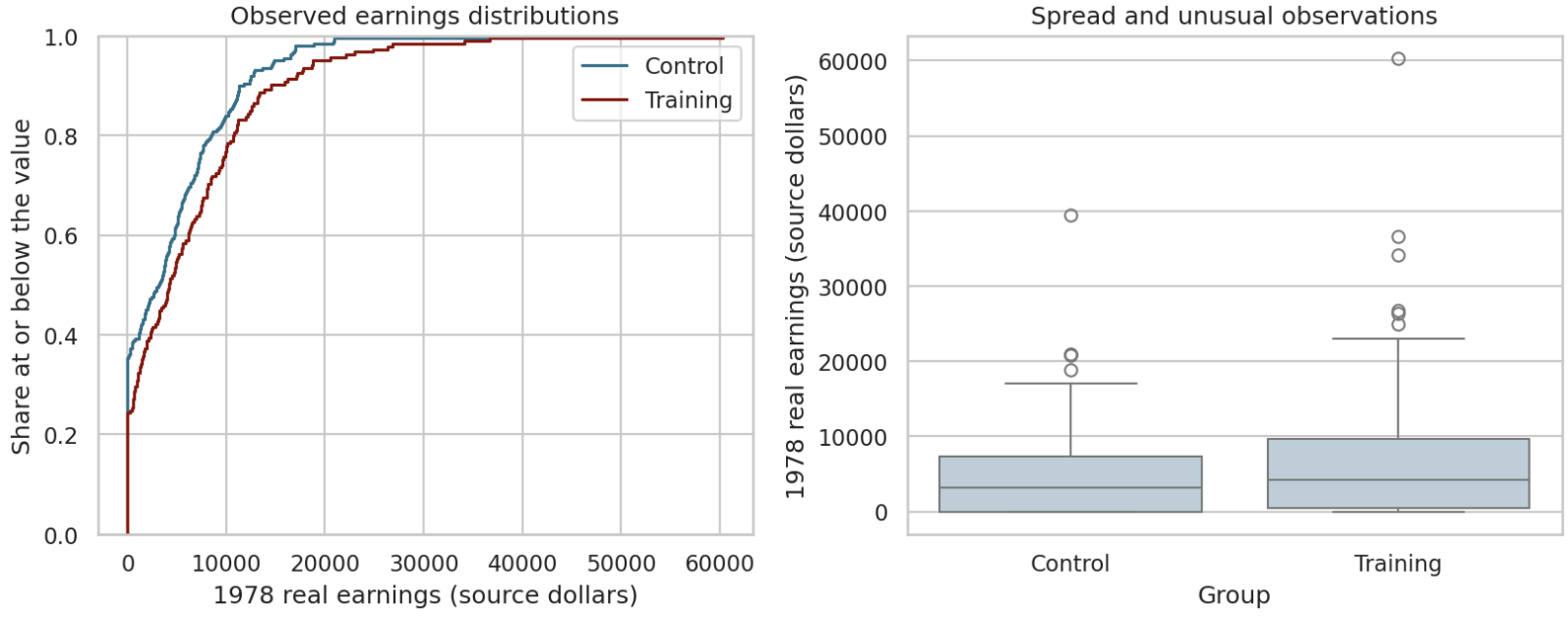
185 training participants

260 controls

Zeros are valid earnings observations, not missing values.

6,349	4,555	1,794	445
Training mean	Control mean	Observed mean gap	Total observed N

The distributions need inspection



> Which features could affect a mean comparison? Discuss for one minute.

2. Sampling variation and confidence intervals

Standard deviation and standard error

SD · individuals

Standard deviation describes how much individual outcomes vary.

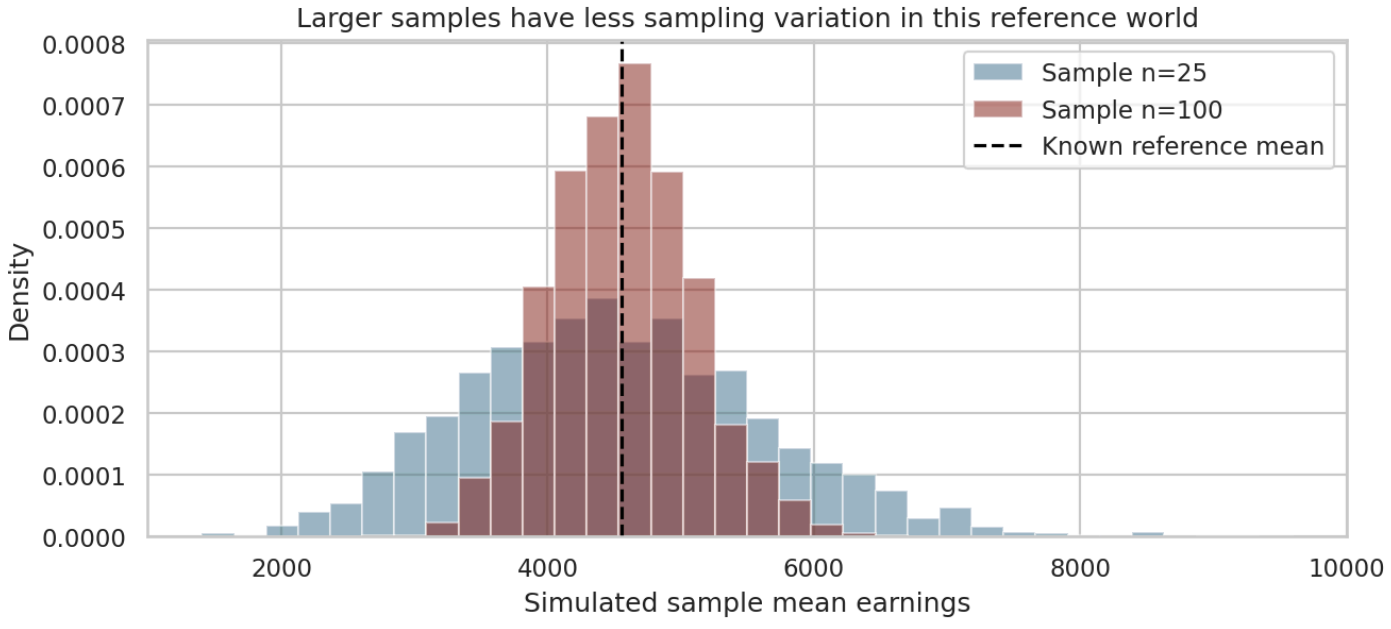
SE · estimates

Standard error describes how an estimator varies across repe
samples under its model.

Independent-group mean contrast: $SE = \sqrt{SD_1^2 / n_1 + SD_0^2 / n_0}$

More observations can reduce SE without making individuals more alike.

A teaching experiment shows sampling variation



Simulated draws from the real control earnings used as a known reference distribution.

Sampling challenge — pairs, 3 minutes

The reference distribution stays the same.

Sample size increases from **25 to 100**.

1. Predict the change in the spread of sample means.
2. Does increasing the sample remove systematic selection bias?
3. Are 2,000 simulated samples the same as observing 2,000 new participants?

Confidence intervals concern estimates

Interval = estimate ± critical value × SE

Repeated procedure

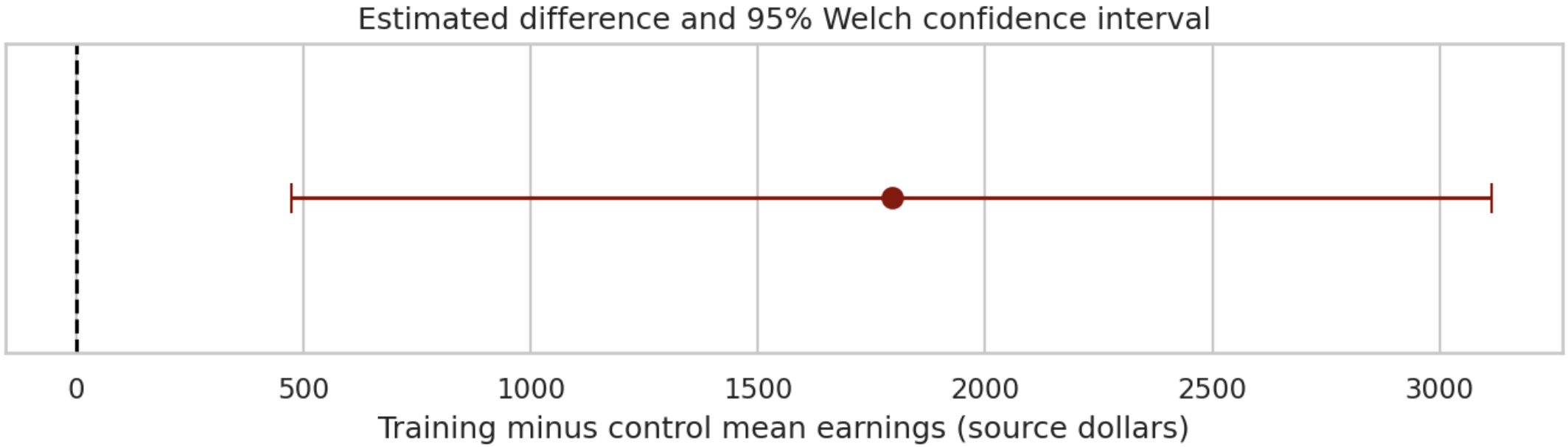
Under its assumptions, a 95% procedure covers the fixed target in approximately 95% of repeated studies.

Observed interval

Report the range, target, method, and assumptions for the in
you calculated.

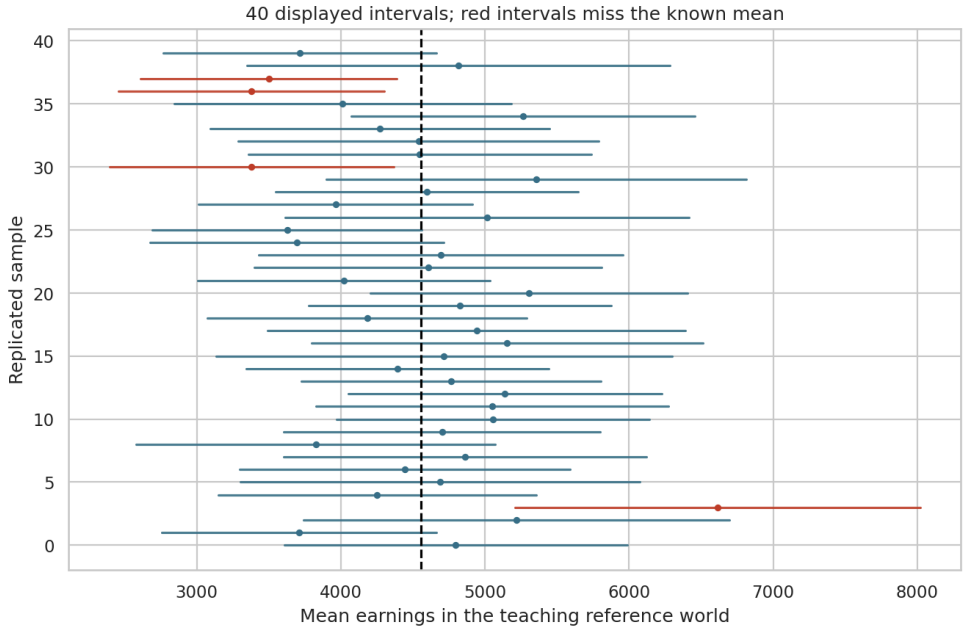
The interval does not describe individual earnings and does not assign a probability to the fixed target.

The real mean difference with uncertainty



Estimate: 1,794. Approximate 95% Welch interval: 474 to 3,115 source dollars.

Repeated intervals illustrate coverage



The dashed line is the known mean of the teaching reference distribution.

Interval interpretation — pairs, 4 minutes

Which claims follow from the earnings interval?

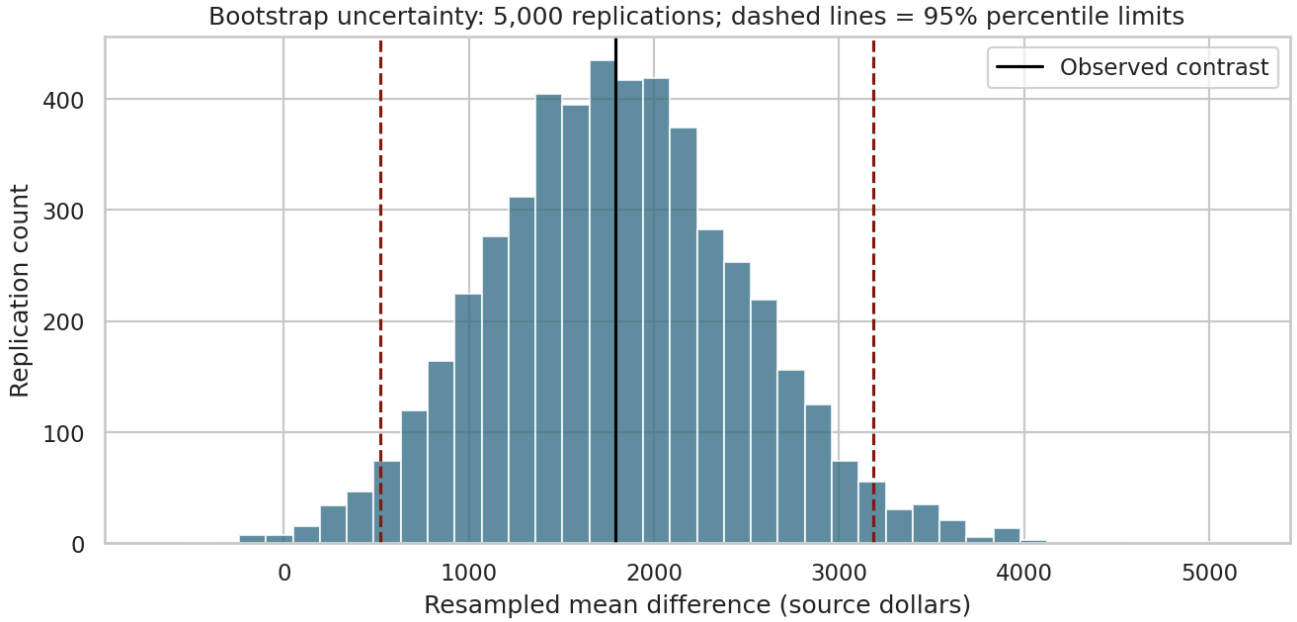
A. Most participants gained between 474 and 3,115 dollars.

B. The estimated mean contrast is positive, with substantial uncertainty about its size.

C. The programme is proven worthwhile in every country.

Explain your choice and repair one incorrect claim.

Bootstrap resampling estimates uncertainty



Resample people within each group with replacement. The contrast varies around the observed estimate.

3. Tests, null models and p-values

Hypothesis test and p-value

The question

Null: the target mean contrast equals zero.

Alternative: the contrast differs from zero.

The two-sided Welch test compares the standardized observed gap with its null reference.

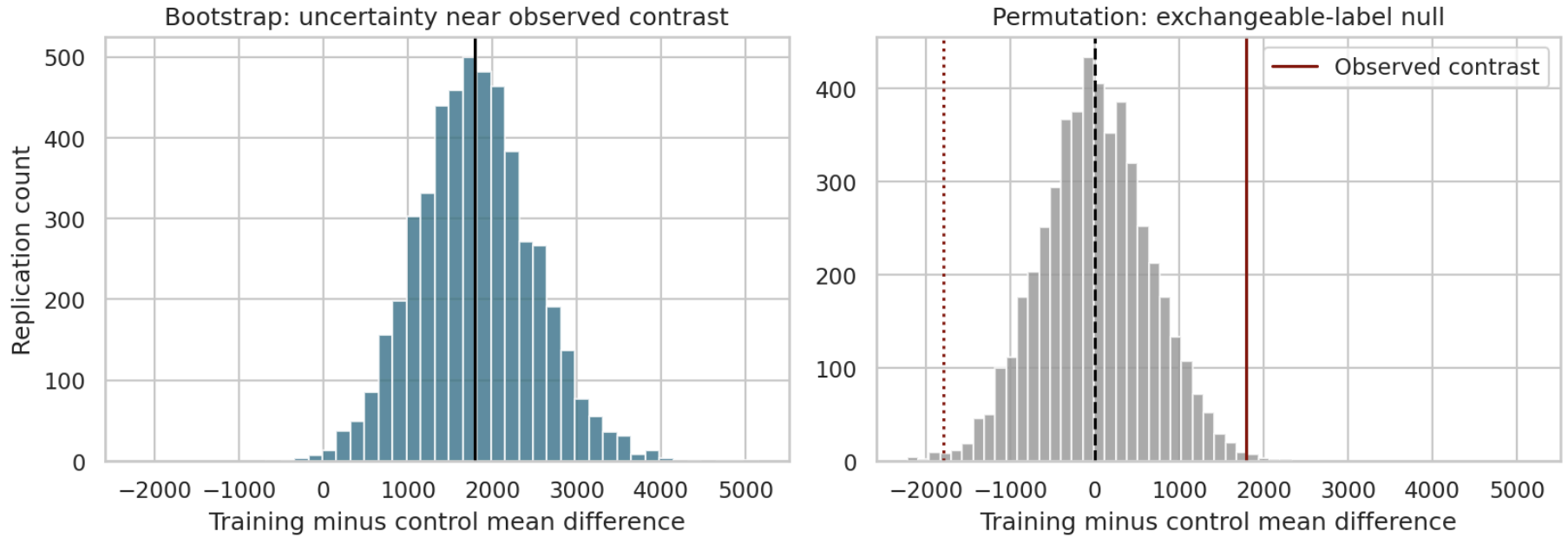
The evidence

Two-sided p = 0.0079

Under the null model, this is the probability of a statistic at lea extreme as the observed one.

> The p-value does not give the probability that the null is true, the size or importance of the contrast, or proof that bias is absent.

A null reference uses a different operation



Bootstrap: uncertainty around the contrast. Shuffled labels: an exchangeable-label null reference.

A result is more than a threshold

For the same two-sided Welch procedure:

The **95% interval excludes zero**, and **p < 0.05**.

That is a statistical conclusion under the procedure's assumptions.

Report the estimate and interval alongside p. A larger p-value would not prove equality or no meanin
effect.

Different methods can yield different intervals or p-values.

4. Effect size and substantive importance

Size, precision and substantive importance

1,794	671	474–3,115	0.273
Mean contrast	Standard error	95% interval	Cohen’s d

Statistical evidence

Two-sided Welch p = 0.0079

Observed sample: 185 training and 260 control participants.

Substantive judgement

Compare the estimate with a justified benchmark, programm
costs, the price basis, and the research setting.

Outcome: 1978 real earnings on the source-dollar scale. Report the estimate and uncertainty, not a software dump.

Judgement challenge — pairs, 4 minutes

Suppose **2,000 source dollars** is an illustrative minimum worthwhile mean gain.

1. Is the estimated contrast above that benchmark?
2. Does the confidence interval establish a gain of at least 2,000?
3. What else would you need before judging programme value?

The benchmark is for discussion; it is not a measured programme cost.

5. Choosing evidence for the research question

Methods follow the variables and design

Independent means	Difference and Welch interval/test	Check units and independence
Repeated people	Within-person differences	Pairing does not make change causal
Two numbers	Scatterplot and Pearson/Spearman	Check shape, unusual values, dependence
Two proportions	Counts and percentage-point gap	Check sample size and outcome definition

> Repeated country-years and clusters need methods that respect dependence.

Two applications answer different questions

Numerical association

Among 260 controls, compare baseline earnings in 1974 and 1975.

Pearson r = 0.675
Spearman ρ = 0.767

Inspect the shape, tied zeros, and unusual values before interpreting the association.

Binary outcome

Positive 1978 earnings: **64.6%** control and **75.7%** training.

Gap = 11.1 percentage points
95% interval: 2.6 to 19.5 points.

Positive earnings do not directly measure employment status

The association is observational. Dichotomizing earnings changes the question and discards information about earnings magnitude.

Evidence choice — pairs, 2 minutes

Choose a starting approach and one limitation:

1. AI reliance and critical-thinking scores from one survey.
2. Changes in the same students' scores over time.
3. Annual poverty observations repeated within provinces.

Explain what would be wrong with treating every row as an independent, representative observation

6. Preparation for Thursday

Bring your current question and a description of your evidence.

Open **W10_RMDS_Inference_Lab.ipynb** in Colab or Jupyter.

The real source data are embedded. No separate download is needed.

You will compare estimates, resampling distributions and interpretations, then select an appropriate form of evidence for your project.

Thursday — Simulation and inference lab

Thursday agenda

1. Inspect the study and estimate the contrast
2. Simulate sampling variation and interval coverage
3. Compare bootstrap and null-reference distributions
4. Apply a method to a different question
5. Write a bounded result and project evidence plan

> Work individually. Use pairs to check interpretations.

Lab steps 1–2: source, estimate, and interval

10 MIN · NOTEBOOK 0–1

Source and first evidence

1. Save a copy and run setup.
2. Check the source and design notes.
3. Inspect the grouped table and distributions.
4. Complete Activity A.

Check: 445 observed participants.

10 MIN · NOTEBOOK 2

Estimate and interval

1. Run the Welch comparison.
2. Identify the estimate, units, sample sizes, SE, interval, an value.
3. Complete Activity B using the 2,000-dollar benchmark.

Lab steps 3–4: sampling and bootstrap uncertainty

10 MIN · NOTEBOOK 3

Sampling and coverage

1. Predict how sample size changes the spread of sample means.
2. Predict whether every 95% interval covers the reference mean.
3. Run the simulation and complete Activity C.

Explain what the teaching reference world represents.

12 MIN · NOTEBOOK 4

Bootstrap uncertainty

1. Resample within each group.
2. Compare bootstrap and Welch intervals.
3. Explain the centre of the bootstrap distribution.
4. Complete Activity D.

> Keep observed participants, sample size, and replication count distinct.

Lab steps 5–6: null reference and application

10 MIN · NOTEBOOK 5

Null-reference experiment

1. Predict the centre after shuffling labels.
2. Compare the null reference with the bootstrap.
3. Explain exchangeability and the design boundary.
4. Complete Activity E in words.

10 MIN · NOTEBOOK 6

Choose one application

A: baseline earnings association and scatterplot.

B: positive-earnings counts, interval, and percentage-point

Complete one response and state its main inferential limita

Lab steps 7–8: result, project choice, and submission

10 MIN · NOTEBOOK 7

Result and project choice

Write an **80–120 word** primary-result paragraph.

For your project, identify the variables, unit, claim, suitable evidence, dependence, and limits.

Check one sentence with a partner.

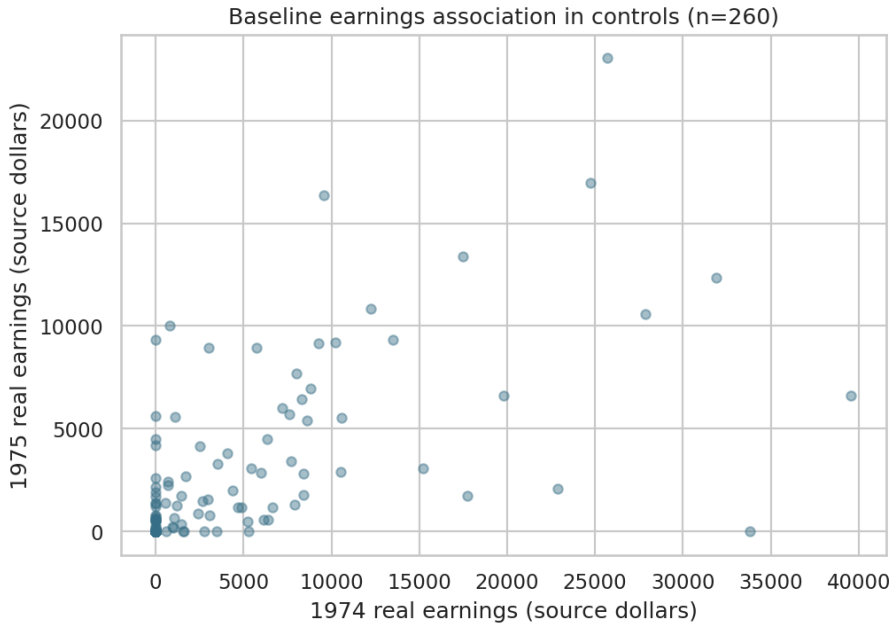
3 MIN · SUBMIT

Save and submit

1. Save the completed notebook.
2. Download the output bundle.
3. Record permitted AI assistance and verification.
4. Follow Cyber Campus instructions.

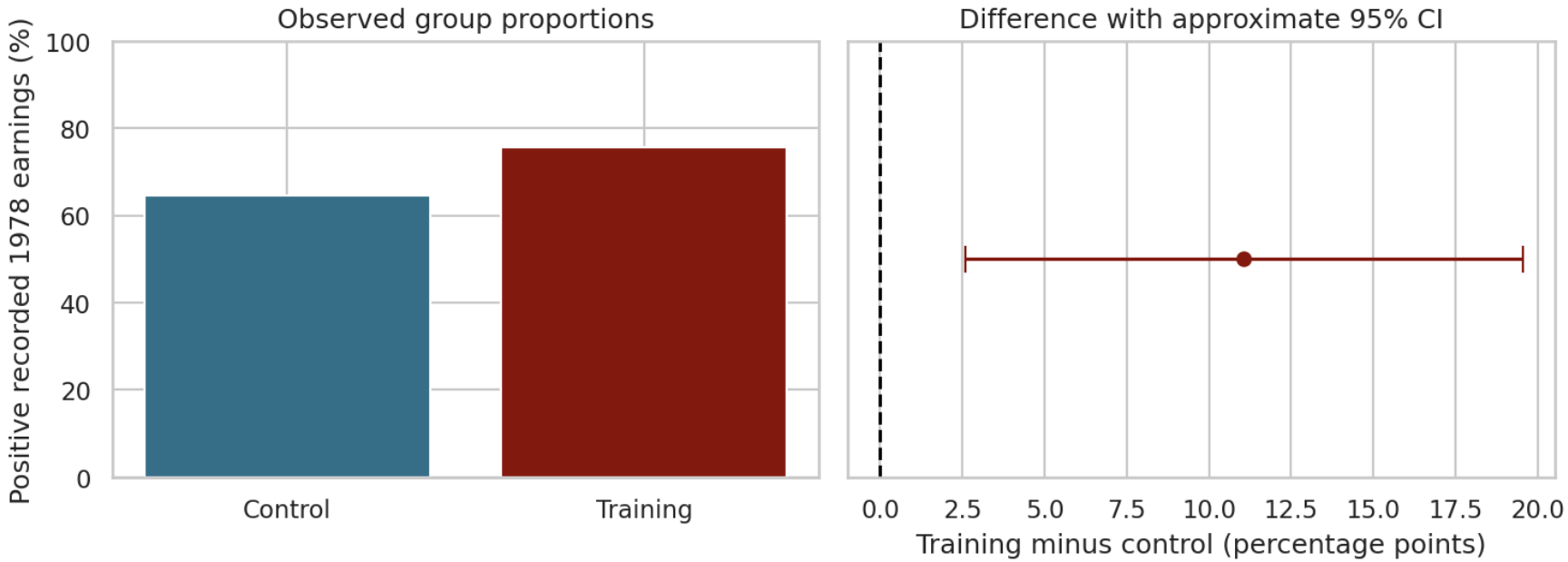
Appendix — reference and stretch materials

Reference: the association plot



Control participants, two pre-intervention earnings variables, n = 260.

Reference: proportions and an interval



Positive recorded earnings, not a directly verified employment rate.

Reference: pairing and method assumptions

Paired observations

When the same person supplies two measurements, analyse the **within-person difference**.

The notebook compares 1975–1978 changes within each group. A before-after change alone does not identify the training effect.

Assumptions to check

Welch: independent records and an appropriate mean target

Bootstrap: resampling unit matches the design.

Permutation: labels are exchangeable under the null.

Association/proportions: dependence, shape, and counts sup
the method.

> A software function cannot validate the research design.

Reference: a research-result checklist

A reader should find:

1. Question and study context
2. Quantity estimated and units
3. Observed sample and comparison
4. Estimate and uncertainty
5. Test result, if appropriate
6. Substantive interpretation and limitations

Report all relevant planned analyses and label exploratory work.

Notebook map and reusable functions

Section / function	Purpose
0–1	Source verification, group table and plots
2 `mean_difference()`	Welch estimate, SE, CI, p and standardized d
3	Sampling variation and coverage demonstration
4 `bootstrap_difference()`	Within-group resampling and percentile interval
5 `shuffle_difference()`	Exchangeable-label null and Monte Carlo p
6	Numerical association or binary comparison
7–9	Reporting, project choice, export and paired extensions

Sources and software documentation

Data and research

- Dehejia data page and NSW files
- LaLonde (1986), *American Economic Review*, 76, 604–620
- Dehejia and Wahba (1999), *JASA*, 94, 1053–1062
- ASA statement on p-values

Software documentation

- SciPy Welch t test
- SciPy Pearson correlation
- SciPy permutation methods

PROVENANCE Original records, retrieval details, and hashes accompany the notebook.